

$$A_v = 20$$

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
FIRST SEMESTER 2024-25

Mid-Semester Examination (Closed Book)

Course No. CE F211

Course Title: Mechanics of Solids

Date: 9th October, 2024

Max. Marks: 60

Duration: 90 Mins

- Q1) Find the force carried in each bar of the hinged equilateral triangle as shown in the Fig. Q1. [12]

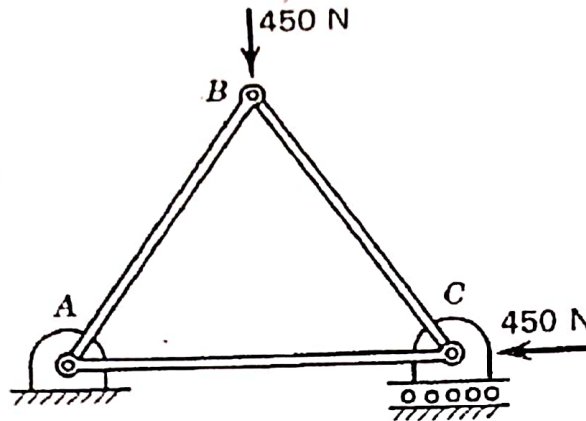


Fig.Q1

- Q2) A rigid bar ACDB is hinged at point A and it is supported by two identical springs having spring constants ks_1 and ks_2 as shown in Fig Q2. Calculate the spring force at point C & D (S_1 & S_2 , respectively) and find the relation between ks_1 and ks_2 . [12+03]

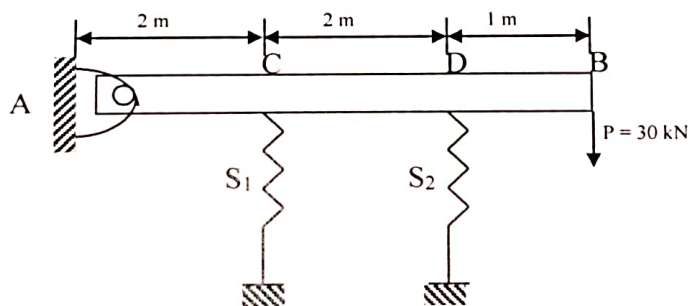


Fig.Q2

- Q3) Calculate the internal forces and moments acting in sections 1 and 2 in the structure as shown in Fig. Q3. Also, derive the expressions of the shear force and bending moment equations. Draw the shear force and bending moment diagrams. [6+6+6]

5/3/27

$$F = kx$$

$$\frac{F}{S_1} = \frac{P}{284}$$

$$F_2 =$$

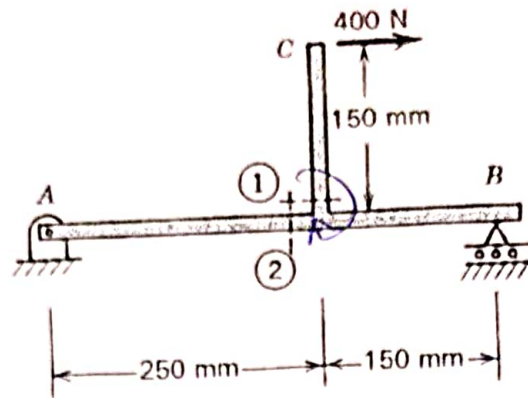


Fig. Q3

Q4) Find the principal stresses magnitudes and their directions if the stress at a point is the sum of the two states of stress illustrated in Fig. Q4. [6+5+4]

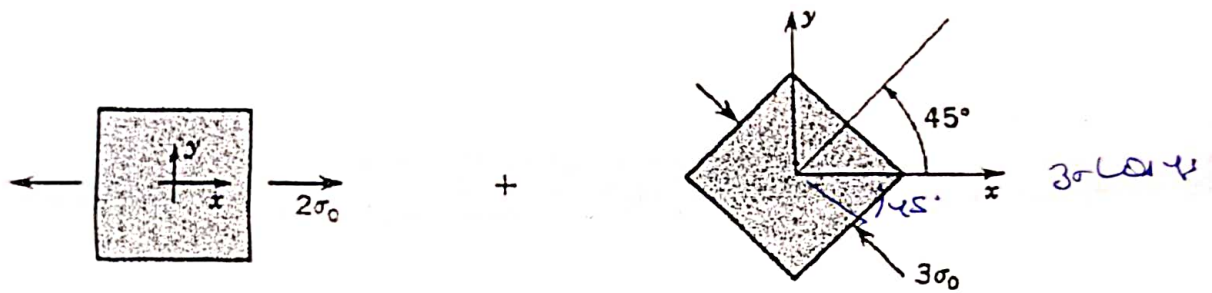


Fig. Q4



Birla Institute of Technology & Science, Pilani

Pilani Campus

I Semester / II Semester / Summer Term 20__ - 20__

Mid-Semester (Regular/Make-Up)

ID No. 2023A2PS1342P Name MANAS CHOUDHARY

Course No. CEF213 Course Title MOS Section No. 12

Instructor's Name Dipendu Room No. 5105 Date 09/10/24

No. of Supplementary Copies
Attached _____

Verified:
Signature of Invigilator: _____

Question No.	Marks Obtained	Student's request for rechecking with remarks	Examiner's remarks
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
Total	(in figures)	(in words)	Examiner's Signature

INSTRUCTIONS

1. Enter all the required details on the cover of every answer booklet.
2. Write on both sides of the sheet in the answer book. Rough work, if any should be done at the bottom of the page. Finally cross out the rough work and draw a horizontal line to separate it from the rest of the material on the page. Also, cross out all blank pages in the answer booklet.
3. Any answer crossed out by the student will not be examined by the examiner.
4. No sheet should be torn from the answer booklet.
5. Mobile phones or any electronic communication/storage device of any kind is prohibited in the examination hall.
6. Use of any unfair means will make the candidate liable to disciplinary action.
7. Student should not leave the examination hall without submitting the answer booklet to invigilator on duty.
8. Student must abide by all the instructions given by the invigilator(s) on duty.

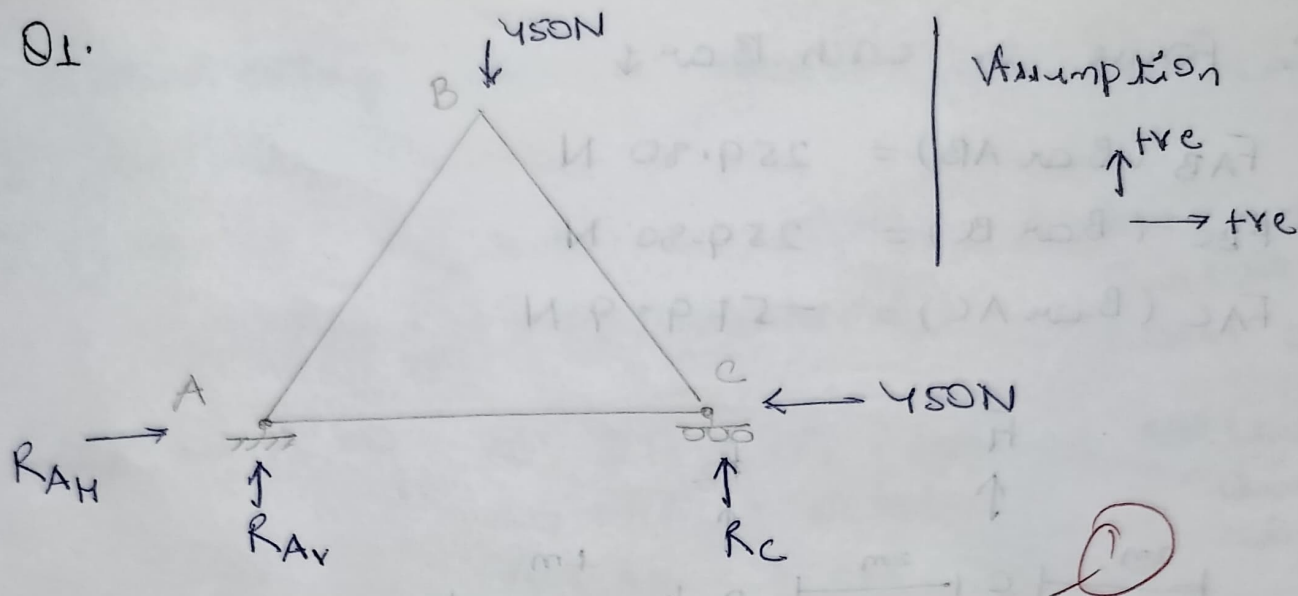
I have carefully read and understood all the instructions.

I do understand that any attempt to use unfair means of any kind in an examination is a serious and punishable offence.

I hereby declare that I will not attempt to do any malpractice in the examination.

M. Anay
Signature of the student

Q1.



- Global force calculation

$$\sum F_x = 0 \Rightarrow R_{AH} - 450 = 0 \Rightarrow R_{AH} = 450 \text{ N} \quad (1)$$

$$\sum F_y = 0 \Rightarrow R_{AV} + R_C = 450 \text{ N} \quad (2)$$

- Joint B:

$$\sum F_x = 0 \Rightarrow F_{BC} = F_{AB} \quad (3)$$

$$\sum F_y = 0 \Rightarrow F_{BC} \cos 30^\circ + F_{AB} \cos 30^\circ + 450 = 0$$

$$2 F_{AB} \cos 30^\circ = 450 \text{ N} \quad (\text{using } (3))$$

$$F_{AB} = \frac{450}{\sqrt{3}} = 259.80 \text{ N} \quad (4)$$

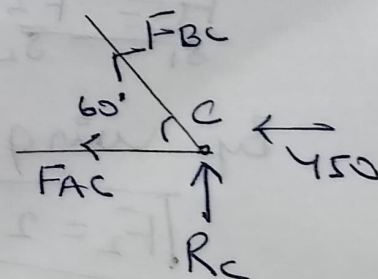
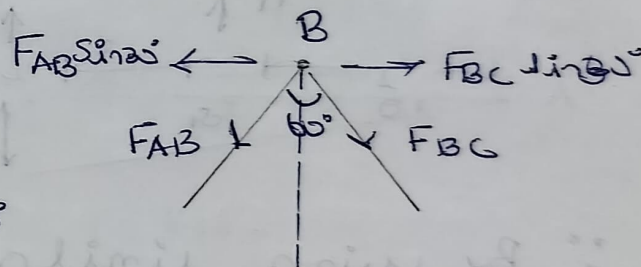
$$\Rightarrow F_{AB} = F_{BC} = 259.80 \text{ N}$$

- Joint C:

$$\sum F_x = 0 \Rightarrow -F_{AC} - F_{BC} \cos 60^\circ - 450 = 0$$

$$F_{AC} = -450 - \frac{259.80}{2}$$

$$F_{AC} = -579.9 \text{ N} \quad (5)$$



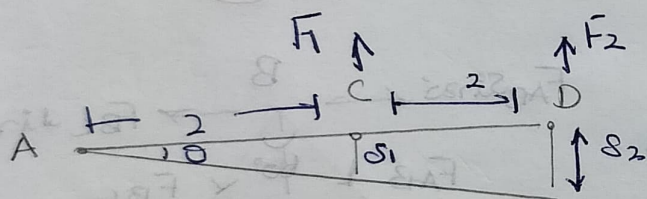
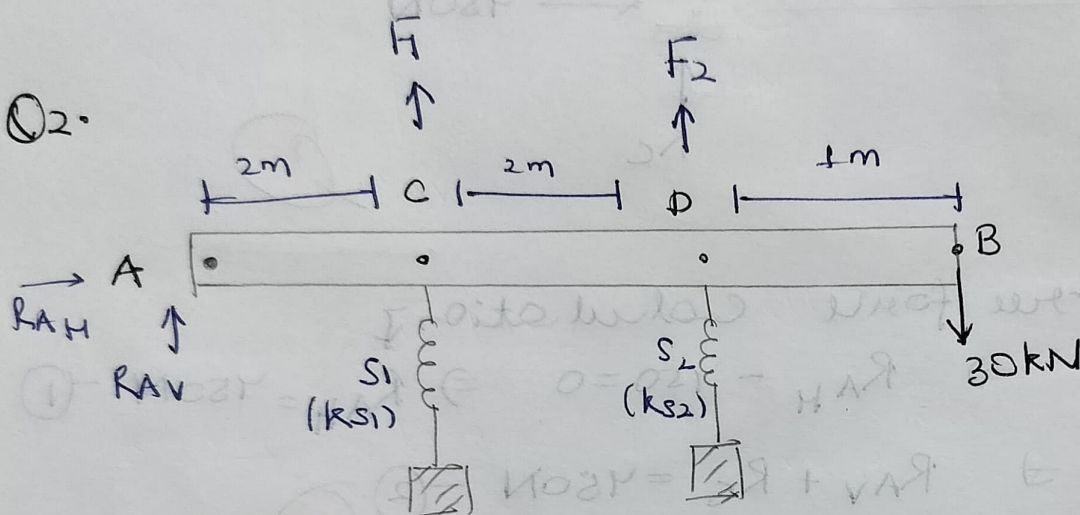
∴ Force in each Bar ↓

$$F_{AB} (\text{Bar AB}) = 259.80 \text{ N}$$

$$F_{BC} (\text{Bar BC}) = 259.80 \text{ N}$$

$$F_{AC} (\text{Bar AC}) = -519.6 \text{ N}$$

Q2.



$s_1 \rightarrow$ deflection in S_1

$s_2 \rightarrow$ deflection in S_2

∴ By using similar Δ

$$\Rightarrow \frac{s_1}{2} = \frac{s_2}{4} \Rightarrow \boxed{s_2 = 2s_1} \quad \text{--- (1)}$$

∴ As springs are identical (similar)
 $(ks_1 = ks_2) \quad \text{--- (2)}$

$$\frac{F_1}{s_1} = \frac{F_2}{s_2}$$

$$\frac{F_1}{s_1} = \frac{F_2}{s_2}$$

$$(F_s = k s)$$

$$(F_s = k/s)$$

$$\frac{F_1}{s_1} = \frac{F_2}{s_2}$$

By using (1) and (2)
 $\boxed{F_2 = 2F_1} \quad \text{--- (3)}$

• Calculating internal force at eqn cond

$$\sum F_x = 0 \Rightarrow R_{AH} = 0$$

$$\sum F_y = 0 \Rightarrow R_{AV} + F + F_2 - 30 = 0 \quad (\text{upward direction is taken as +ve})$$
$$R_{AV} + F + F_2 = 30 \text{ kN} \quad \text{--- (4)}$$

$$\sum M_z \text{ at A} = 0 \Rightarrow 2F + 4F_2 - 30 \times 5 = 0 \quad (\text{clockwise direction is +ve})$$
$$2F + 4F_2 = 150 \text{ kN}$$
$$F + 2F_2 = 75 \text{ kN} \quad \text{--- (5)}$$

$\therefore$ using (5) & (3) $F + 2(2F) = 75$

$$F = 15 \text{ kN} \quad \text{--- (6)} \quad (2)$$

$\therefore$ using (6) & (3) $F_2 = 30 \text{ kN} \quad \text{--- (7)} \quad (2)$

$\therefore$ Put values of F & F_2 in (4)

$$R_{AV} + 30 + 15 = 30 \Rightarrow R_{AV} = -15 \text{ kN} \quad \text{--- (2)}$$

$\therefore$ spring force at C (F) = 15 kN

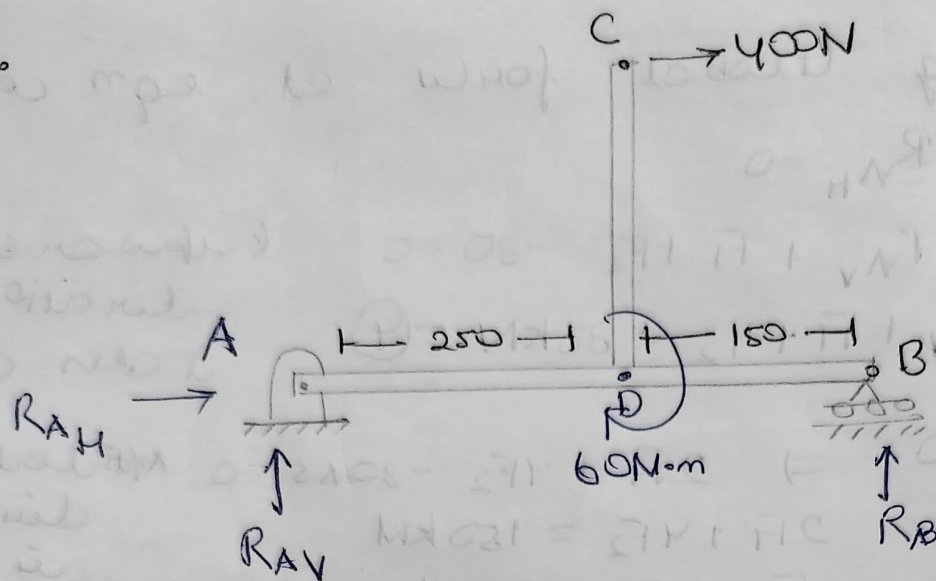
spring force at D (F_2) = 30 kN

$$kS_1 = kS_2 \quad (3)$$

$$R_{AV} = -15 \text{ kN}$$

$$R_{AH} = 0 \text{ kN}$$

Q3.



→ Calculating Reaction forces,

* Assumption, $\uparrow +ve$, $\rightarrow +ve$, $\curvearrowright +ve$

$$\sum F_x = 0 \Rightarrow R_{AH} + 400 = 0 \Rightarrow R_{AH} = -400N$$

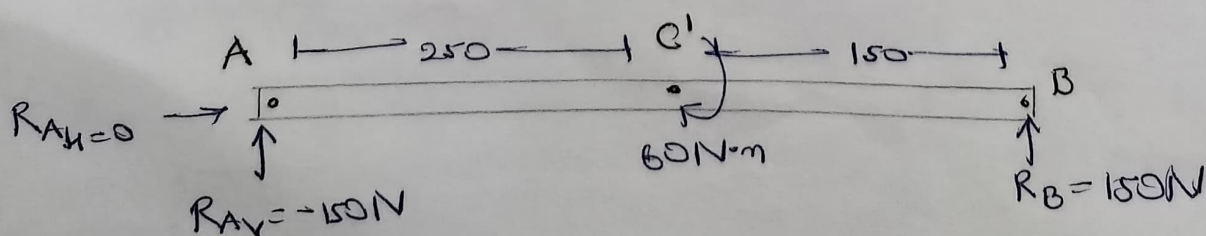
$$\sum F_y = 0 \Rightarrow R_{AV} + R_B = 0 \quad \text{--- (2)}$$

$$\sum M_z \text{ at A} = 0 \Rightarrow -60 + R_B(0.4) = 0$$

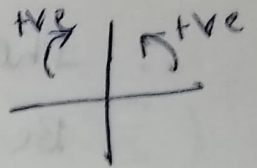
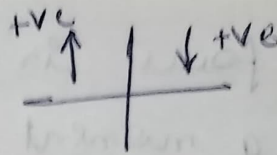
$$R_B = \frac{60}{0.4} = 150N \quad \text{--- (1)}$$

$$\therefore \text{ Put (1) in (2) } \Rightarrow R_{AV} = -150N \quad \text{--- (2)}$$

$\therefore$ Here I have converted 400N force to moment of 60 N·m

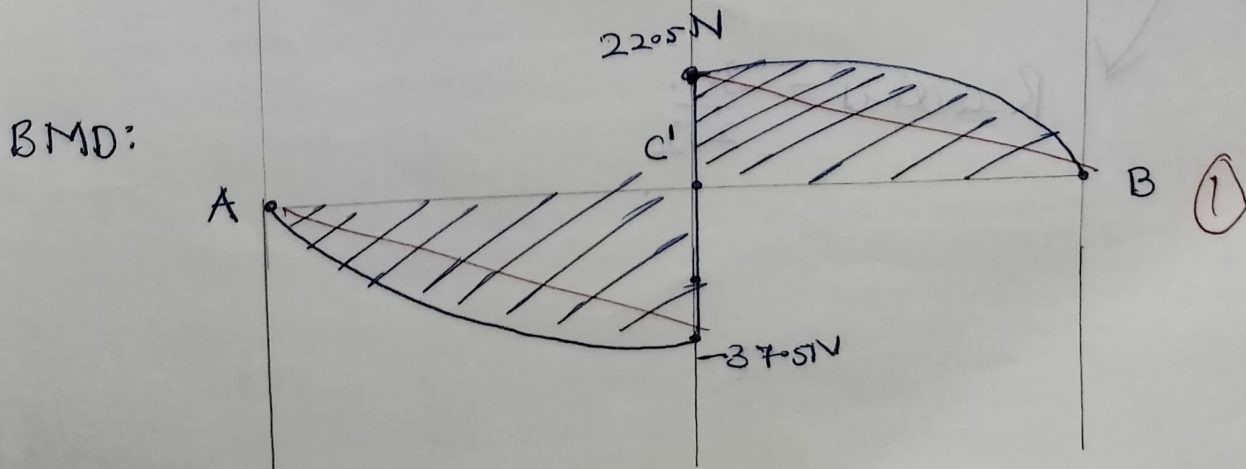
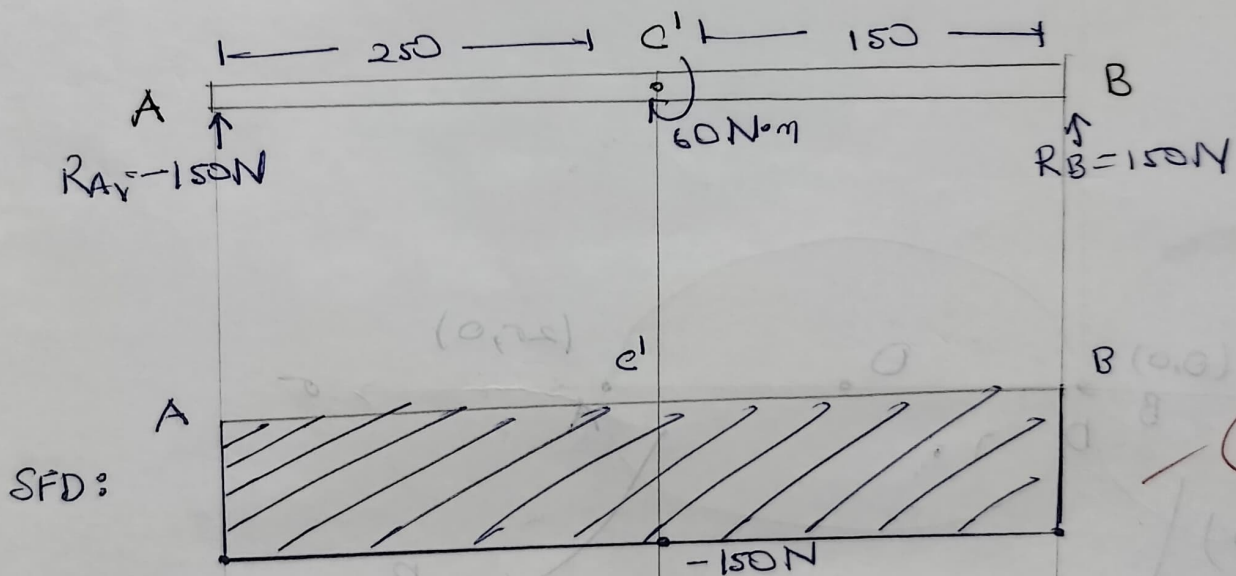


* Sign Convention —



By using Section Method,

- Shear force b/w A-C' = -150 N (2) + (2)
- Shear force b/w C'-B = -150 N
- Bending moment at A = $60 \text{ N}\cdot\text{m}$
- Bending moment at B = $60 + (-150)(0.4) = 0 \text{ N}\cdot\text{m}$
- Bending moment at C' (left) = $\frac{-150 \times 250}{1000} = -37.5 \text{ N}\cdot\text{m}$ (4)
- Bending moment at C' (right) = $\frac{-150 \times 250}{1000} + 60 = 22.5 \text{ N}\cdot\text{m}$ (4)



$\therefore$ Shear force in section ① & ② = -150N
 Bending moment in ① = $-37.5\text{N}\cdot\text{m}$
 Bending moment in ② = $60\text{N}\cdot\text{m}$

Q4.



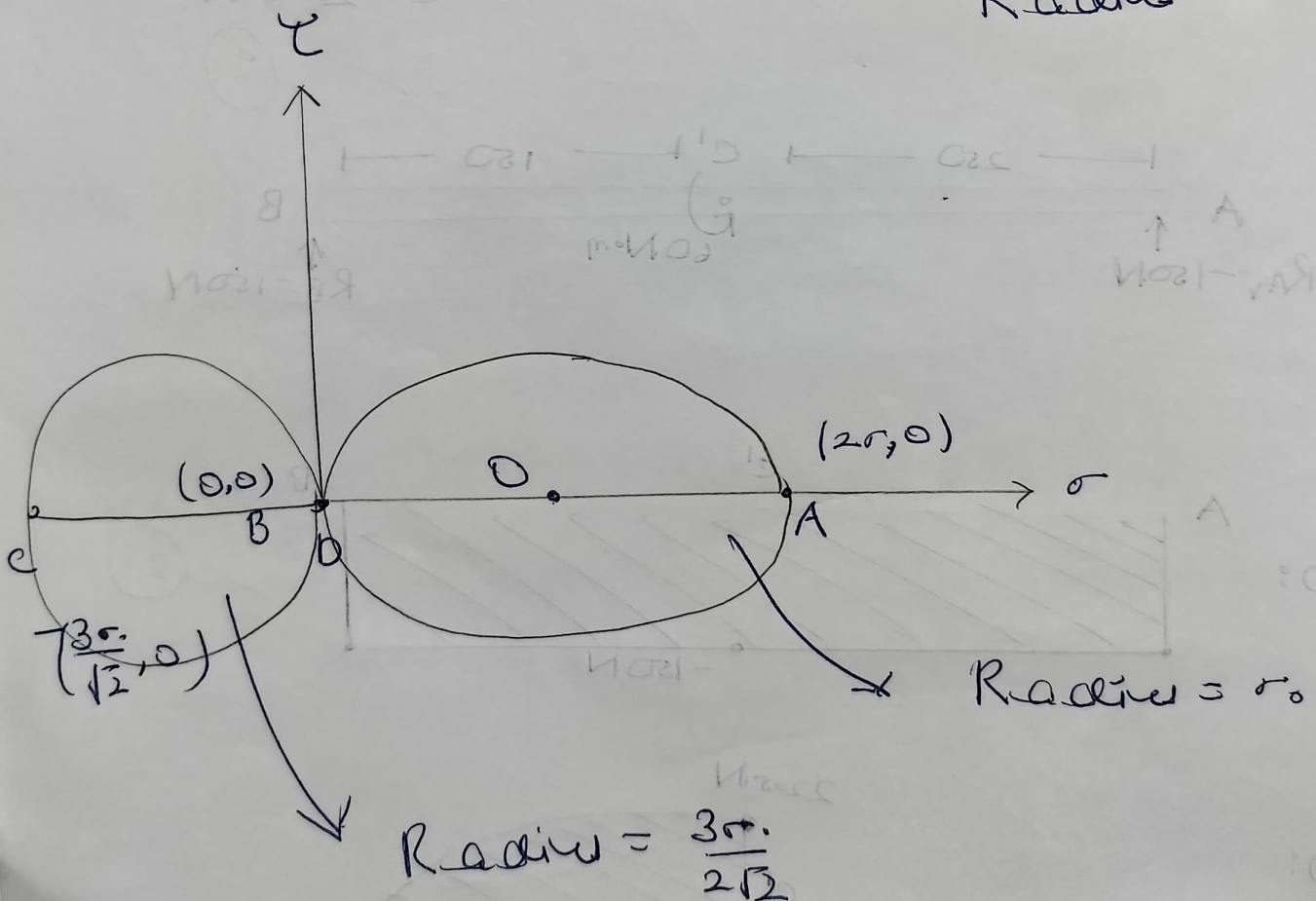
$$A = (\sigma_{xx}, -\tau_{xy})$$

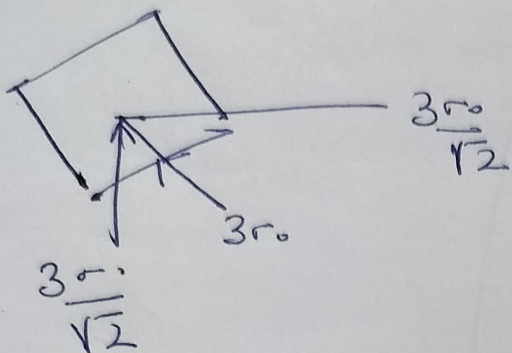
$$B = (\sigma_{yy}, \tau_{xy})$$

$$A = (2\sigma_x, 0)$$

$$B = (0, 0)$$

$\downarrow$
 Diameter = 2σ
 Radius = σ





$$C = \left(\frac{3r_0}{\sqrt{2}}, \frac{3r_0}{\sqrt{2}} \right)$$

$$C = \left(-\frac{3r_0}{\sqrt{2}}, 0 \right)$$

$$D = \left(-0, 0 \right)$$